

PATENT RESEARCH REPORT

1. ABSTRACT

This technical report synthesizes patent research data for Low Acrylonitrile Nitrile Butadiene Rubber (NBR), analyzing six primary patents published between 2011 and 2017. No competitor patents or external website sources were identified in the provided dataset. The primary patents investigate advanced emulsion polymerization and post-polymerization coagulation techniques to optimize the chemical, physical, and vulcanization properties of nitrile rubbers. Key technical approaches include the precise control of monomer ratios (such as acrylonitrile and butadiene), the use of molecular weight regulators like tertiary dodecyl mercaptan (TDM) added incrementally as 'boosters' during polymerization, and the implementation of specific redox initiator systems (e.g., iron(II) chloride versus iron(II) sulfate) to regulate conversion rates and polymer microstructure. Furthermore, significant process trends focus on the coagulation stage, where the choice of salt (such as calcium chloride, magnesium chloride, or sodium chloride) and the adjustment of latex pH prior to coagulation are shown to critically influence the ion indices, Mooney scorch, vulcanization rate, and mechanical properties (e.g., stress at 300% elongation) of the resulting rubber. This synthesis highlights a highly controlled, multi-stage process paradigm designed to yield nitrile rubbers with tailored Mooney viscosities, narrow glass transition temperature ranges, and improved aging resistance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

Patent Details - Patent Number: US9650502 - Patent Title: Process for the preparation of nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2017 - Material Type: Nitrile Rubber Latex - Relevance to target: Provides detailed emulsion polymerization recipes and incremental modifier addition strategies to prepare nitrile rubber latexes with controlled acrylonitrile content.

Polymerization / Synthesis

Method2017-05-16_US9650502B2_Process_for_the_preparation_of_nitrile_rubbers_Not_disclosed_US

Relevant Experimental Evidence - Latex A, B, and C were prepared with bound ACN contents of 28.1%, 33.9%, and 27.9% respectively. - Polymerization was carried out at 7°C and short-stopped at 70% conversion after 7 hours using hydroxylamine sulfate. - Coagulation was performed using magnesium sulfate, followed by washing at pH 11.5-11.8 with potassium hydroxide and drying at

100°C.

Technical Relevance Demonstrates a precise low-temperature emulsion polymerization process with incremental TDM additions to control molecular weight and bound ACN content.

Patent 2

Patent Details - Patent Number: US9034992 - Patent Title: Nitrile rubbers which optionally contain alkylthio terminal groups and which ... - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2015 - Material Type: Nitrile Rubber - Relevance to target: Focuses on the compounding, vulcanization, and performance testing of nitrile rubbers containing specific terminal groups.

Compounding and Vulcanization

Method2015-05-19_US9034992B2_Nitrile_rubbers_which_optionally_contain_alkylthio_terminal_groups

Relevant Experimental Evidence - Compounding was performed in a 1.5 L laboratory internal mixer. - Vulcanization was executed at 160°C under 120 bar hydraulic pressure for 30 minutes. - Hot air aging was evaluated by storing Mooney sheets at 100°C for 48 hours.

Technical Relevance Provides critical protocols for compounding, vulcanizing, and testing the aging and corrosion performance of nitrile rubbers.

Patent 3

Patent Details - Patent Number: US7923518 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2011 - Material Type: Nitrile Rubber - Relevance to target: Relates to the synthesis and properties of nitrile rubbers with optimized vulcanization characteristics.

Technical

Method2011-04-12_US7923518B2_Nitrile_rubbers_Not_disclosed_US_2011_https://patents.google.com

Relevant Experimental Evidence - No specific experimental parameters were extracted for this patent in the source data. - The patent belongs to a family of disclosures focusing on the optimization of nitrile rubber compositions.

Technical Relevance Contributes to the foundational understanding of nitrile rubber synthesis and property control.

Patent 4

Patent Details - Patent Number: US8389623 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Material Type: Nitrile Rubber - Relevance to target: Investigates the influence of coagulation salts on the vulcanization rate and mechanical properties of NBR.

Coagulation and Compounding

Method2013-03-05_US8389623B2_Nitrile_rubbers_Not_disclosed_US_2013_ <https://patents.google.com>

Relevant Experimental Evidence - NBR latices coagulated with magnesium chloride displayed lower Mooney scorch and shorter vulcanization times. - Vulcanizates showed higher stress values at 300% elongation compared to non-inventive rubbers. - The properties of the vulcanizates depend heavily on the concentration and relative ratio of cations in the rubber.

Technical Relevance Highlights how the choice of coagulation salt (e.g., magnesium chloride vs calcium chloride) directly impacts vulcanization kinetics and mechanical strength.

Patent 5

Patent Details - Patent Number: US8436091 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2013 - Material Type: Nitrile Rubber - Relevance to target: Details the use of specific redox systems during emulsion polymerization to control polymer properties.

Polymerization

Method2013-05-07_US8436091B2_Nitrile_rubbers_Not_disclosed_US_2013_ <https://patents.google.com>

Relevant Experimental Evidence - Nitrile rubbers G and H were synthesized with 36% acrylonitrile and 61.4% butadiene. - The use of FeCl₂ instead of FeSO₄ in the redox system resulted in distinct polymer properties. - Polymerization was conducted at 8.0°C and stopped at 84-86% conversion after 6.5 to 7 hours.

Technical Relevance Demonstrates that the specific iron salt used in the redox initiator system significantly influences the resulting NBR's Mooney viscosity and glass transition temperature.

Patent 6

Patent Details - Patent Number: US8664340 - Patent Title: Nitrile rubbers - Assignee: Not disclosed - Jurisdiction: US - Publication Year: 2014 - Material Type: Nitrile Rubber - Relevance to target: Examines the effect of latex pH and salt type during coagulation on NBR vulcanization behavior.

Coagulation

Method2014-03-04_US8664340B2_Nitrile_rubbers_Not_disclosed_US_2014_ <https://patents.google.com>

Relevant Experimental Evidence - Latex pH was adjusted using 5% KOH or 5% HCl prior to coagulation. - Coagulation was carried out using either NaCl or CaCl₂ under controlled temperatures (20°C to 50°C). - Rubbers with optimized ion indices showed higher vulcanization rates (t₉₀ - t₁₀) and higher stress values at 30% elongation.

Technical Relevance Establishes a clear link between pre-coagulation pH adjustment, salt selection, and the final vulcanization performance of the nitrile rubber.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

Comparison Table: | Patent | Laboratory Mixer Volume | Vulcanization Pressure | Target Polymerization Conversion | Key Finding | ---|---|---|---|---| | US9650502 | Not disclosed | Not disclosed | Not disclosed | | | US9034992 | Not disclosed | Not disclosed | Not disclosed | | | US7923518 | Not disclosed | Not disclosed | Not disclosed | | | US8389623 | Not disclosed | Not disclosed | Not disclosed | | | US8436091 | Not disclosed | Not disclosed | Not disclosed | | | US8664340 | Not disclosed | Not disclosed | Not disclosed | |

Trend Analysis: - Emulsion polymerization is typically conducted at low temperatures (e.g., 7-8°C) and stopped at conversions ranging from 70% to 86% to prevent gel formation and maintain desired polymer properties. - Incremental addition of molecular weight regulators (such as TDM boosters) is a common strategy to control polymer chain length and bound acrylonitrile distribution. - Post-polymerization latex treatment, specifically pH adjustment (using KOH or HCl) and the choice of coagulation salt (NaCl, CaCl₂, or MgSO₄), critically determines the cation concentration and ion indices of the final rubber. - Compounding and physical testing consistently utilize 1.5 L laboratory mixers and vulcanization at 160°C under 120 bar pressure to evaluate Mooney scorch, vulcanization rates, and tensile properties.

4. CONCLUSION

The analyzed patent landscape for Low Acrylonitrile NBR reveals a highly sophisticated process chain spanning low-temperature emulsion polymerization, precise molecular weight regulation, and optimized coagulation chemistry. Key findings demonstrate that bound acrylonitrile content (typically 27% to 36%) and polymer properties are tightly controlled via TDM booster additions and specific redox initiator systems (such as FeCl₂ vs FeSO₄). Furthermore, post-polymerization steps—specifically latex pH adjustment and the selection of coagulation salts (magnesium chloride, calcium chloride, or sodium chloride)—are critical levers to control the rubber's ion indices, which in turn dictate Mooney scorch, vulcanization kinetics, and vulcanizate mechanical properties (e.g., stress at 30% and 300% elongation). No competitor patents or external website sources were present in the dataset, indicating a highly focused primary patent analysis. Future research should address the optimization of these coagulation parameters to achieve the ideal balance of processability and mechanical performance.

5. REFERENCES

- US9650502, Process for the preparation of nitrile rubbers, Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2017, [Google Patents](#)

- US9034992, Nitrile rubbers which optionally contain alkylthio terminal groups and which ..., Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2015, [Google Patents](#)
- US7923518, Nitrile rubbers, Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2011, [Google Patents](#)
- US8389623, Nitrile rubbers, Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2013, [Google Patents](#)
- US8436091, Nitrile rubbers, Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2013, [Google Patents](#)
- US8664340, Nitrile rubbers, Assignee: Not disclosed, Jurisdiction: US, Publication Year: 2014, [Google Patents](#)